7005SCN INDIVIDUAL RESEARCH PROJECT

CW1 Project Proposal (Proforma)

Word Count: 1000 words ±10%

**Current words on template 351. Do not include these in word count.*

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Course	Individual Research Project
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Ethics Application Status

Tick ONE option:

☐Approved

Ethics ID:

Submitted, under review ✅,

☐Not yet submitted

1. Research Question/Problem Statement

Clearly define your research problem or question (or hypothesis), situating it within its relevant domain or context. Explain why this problem is important, what gap or issue it addresses and why it is worthy of investigation.

Employee turnover will cost the business: recruitment budget, management time and organisational knowledge will be spent on replacing an experienced employee. There has been a growing interest in using machine learning to identify employees at risk of departure, and Alqahtani et al. (2024) report that the accuracy of such predictions is often high. But even if accurate, this is not enough to assist an HR manager in determining whether to take action, and if he takes it, whether he is correct, because he will need to explain it if he is challenged. Risk score is not helpful unless there is a reason. Explainable AI (XAI) is thus an essential component of responsible HR analytics, but many existing reports give importance as a side-effect instead of testing the stability and usefulness of explanations. Additionally, Rudin (2019) asserts that an interpretable model could be better suited to decisions with an impact than post-hoc explanation of a black-box, which is rarely evaluated in this context.

RQ1. Which employee characteristics are the strongest predictors of attrition, and how consistent are they across models and explanation methods?

RQ2. How does an intrinsically interpretable model compare with a black-box model on predictive performance, and what is the cost of interpretability?

RQ3. To what extent can XAI outputs be translated into actionable, defensible guidance for HR managers?

2. Intended Users/Stakeholders and Needs

Identify the key users, stakeholders or organisations who will benefit from your project. Explain their needs, challenges and describe how your proposed research aims to address or respond to these needs.

People analytics for retention and HR managers are the main users. They need more than a risk list, they need to understand what is driving each prediction and if it is controllable within the organisation, and the level of confidence of the model for each prediction to ensure that retention budget is spent wisely. Line managers are secondary users, not getting raw model outputs but rather explanations showing the results; governance and compliance staff is required to ensure that the retention decisions are not discriminatory and can be explained. The project responds by generating not only prediction results, but evidence of the quality of the explanations, making a clear comparison between interpretable and black-box models, and providing suggestions on how far explanations can go beyond mere description of model behaviour toward supporting intervention.

3. Project Scope, Deliverables and Expected Outcomes

Outline the scope and boundaries of your project, including what will and will not be included. Describe the artefact, system, model or analysis you intend to produce, the key deliverables, and the expected outcomes, making clear how these align with and address the research problem.

Scope: binary attrition classification on the IBM HR Analytics dataset; interpretable models (logistic regression, decision tree, explainable boosting machine) vs. black-box models (random forest, gradient boosting); SHAP and LIME explanations; stability and agreement analysis; and fairness checks by gender and by age group. These are some of the areas that are outside the scope: live deployment; collection of new employee data; causal estimation of intervention effects; time-to-event modelling; recommendations for identified individuals.

Deliverables: A reproducible Python pipeline for the processing, handling of imbalances, training and evaluation, a comparative results table, global and local explanations per model, a quantitative analysis of explanations agreement and stability, a guidance note for practitioners, and a dissertation with public code repository.

Expected outcomes: identification of the most successful and stable predictors of attrition, evidence as to whether interpretability comes at a cost of meaningful predictive performance on this dataset and a considered evaluation of what XAI can inform HR managers. These directly answer RQ1-RQ3.

4. Research Methodology

Explain how your research will be conducted by outlining your overall approach (e.g., experimental, case study, design-based, analytical). Describe the methods, techniques, tools, technologies and data sources you will use, including how data will be collected and analysed where relevant. Provide justification for your methodological choices.

It is quantitative and experimental approach and it employs an existing public dataset. No participants are recruited and no personal data are processed: The HR Analytics dataset (IBM, 2016) is a fictional database of 1470 records designed by the data scientists at IBM to mirror real HR workflows, but it does not contain any personal employee information, which leads to reduced external validity and the absence of privacy risk.

Preprocessing includes removal of constant and identifier columns, encoding of categorical variables and scaling. Class weighting and SMOTE (Chawla et al., 2002) are compared with the target being imbalanced at about 16% : only resampling would be done within the folds of the cross validation to prevent optimistic bias. Models are tested on the stratified five-fold cross-validated and on the held-out test sets, using ROC-AUC, PR-AUC, recall and F1 metrics instead of accuracy, because the latter is 84% for a majority class predictor, and 0 for one that predicts nothing.

For global and local attribution explanations, SHAP (Lundberg & Lee, 2017) is used, and local comparison explanations are explained using LIME (Ribeiro et al., 2016). Instead of taking the explanations at face value, the study assesses them: agreement of SHAP and LIME, local explanations' stability across resampling, small perturbations and agreement with the interpretable models' coefficients. This design directly implements Rudin's (2019) critique and assesses the necessity of post-hoc explanation. The implementation is in Python, scikit-learn, XGBoost, imbalanced-learn, SHAP and LIME.

5. Initial Literature Review

Provide a concise but critical review of 3–5 academic sources. Summarise key findings and approaches, compare and synthesise perspectives, identify gaps or limitations in the existing literature and explain how your proposed project will address these gaps.

Alqahtani et al. (2024) do an update of the literature from 2019-2024 on machine learning for prediction of attrition, and their results reveal that there is a consistent strong performance by both tree ensembles and neural approaches, but there is also split practice: some studies differ in their datasets, some handle imbalance differently, some use imbalance-specific metrics, and many report accuracy on an imbalanced target, which over-inflates apparent success. They present the synthesis, which shows that prediction is possible but comparability and interpretation is not.

SHAP unifies all additive attribution methods under a single paradigm—SHAP—which guarantees consistency that the previous importance measurement methods did not. SHAP has since become the standard for applied HR analysis.SHAP is now the standard for applied HR analysis, and has been merged under a single paradigm by Lundberg and Lee (2017) that guarantees consistency that previous importance measurement methods did not. Ribeiro et al. (2016) had previously proposed LIME, where they were approximating a model

locally with a surrogate. They both describe model behavior and not the world, and LIME is sensitive to sampling, meaning that two runs can provide different explanations for one prediction – a characteristic that is seldom reported in applied studies.u

Rudin (2019) calls this literature into question, specifically stating that post-hoc explanations of black boxes are only approximations, and that interpretable models should be preferred if they are comparable in performance in high-stakes contexts. Given it's structured tabular data of this size, her position can be easily imagined, but HR attrition research rarely challenges it.

The field can thus anticipate attrition and create explanations but it rarely assesses whether the explanations are stable, internally consistent, or more convincing than if they had been put forward using the transparent model. The project does not assume that explanation quality will automatically occur, but rather measures it.

6. Project Planning and Management

Provide a structured and feasible plan explaining how your project will be carried out over time. This should include a clear timeline of key stages, tasks and milestones, demonstrating sequencing and dependencies. You should also explain the feasibility of the project in terms of time, resources and scope and identify potential risks along with appropriate mitigation strategies.

Weeks 1-3: review literature, gather data and perform exploratory analysis. Week 4-6: Preprocessing pipeline, comparing imbalance strategies, and stable interpretable models. Weeks 7-9: black-box models, tuning and complete performance comparison. Week 10-12: SHAP and LIME explanations and stability, agreement and fairness analyses. Weeks 13-15: Guidance note and Dissertation Writing, along with the repository. The most critical dependency is the imbalance strategy; if it is later changed, then previous results would not be valid.

Feasibility: high - small data set, public, clean and all the tooling is open source, and the work is performed on a standard laptop. Principal risks include weak performance of interpretable model (which is acceptable and not a failure, and will be reported accordingly), unstable LIME explanations (which will be mitigated by setting random seeds and explicitly reporting on the variation in LIME results), and overclaiming of the correlational results (which will be mitigated by reporting on the correlation in a synthetic dataset, rather than determining drivers of turnover).

7. References (APA Style)

List all references using APA format.

Alqahtani, H., Almagrabi, H., & Alharbi, A. (2024). Employee attrition prediction using machine learning models: A review paper. International Journal of Artificial

Intelligence & Applications, 15(2), 23-49. https://doi.org/10.5121/ijaia.2024.15202

Chawla, N. V., Bowyer, K. W., Hall, L. O., & Kegelmeyer, W. P. (2002). SMOTE: Synthetic minority over-sampling technique. Journal of Artificial Intelligence Research, 16, 321-357. https://doi.org/10.1613/jair.953

IBM. (2016). IBM HR Analytics employee attrition and performance [Data set]. Kaggle. https://www.kaggle.com/datasets/pavansubhasht/ibm-hr-analytics-attrition-dataset

Lundberg, S. M., & Lee, S.-I. (2017). A unified approach to interpreting model predictions. Advances in Neural Information Processing Systems, 30, 4765-4774.

Ribeiro, M. T., Singh, S., & Guestrin, C. (2016). "Why should I trust you?": Explaining the predictions of any classifier. Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, 1135-1144. https://doi.org/10.1145/2939672.2939778

Rudin, C. (2019). Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead. Nature Machine Intelligence, 1(5), 206-215. https://doi.org/10.1038/s42256-019-0048-x

AI Usage Declaration

Explain if and how AI tools were used.

Generative AI (Claude, Anthropic) was used in a supporting role only. The research topic, questions and study design are the author's own. AI assistance was used to verify citation details against published sources and to improve conciseness within the word limit. All wording was reviewed and revised by the author, every source cited was read independently, and no AI tool generated results, data or analysis.